Gas, totals and mass flow

Cold gas (stations 0-3, 13-19): γ_c = 1.4, cp_c = 1004.5 J/(kg K), R_c = 287 J/(kg K). Hot gas (4-9): γ_t = 1.3, cp_t = 1239 J/(kg K), R_t = cp_t (γ_t-1)/γ_t = 285.9 J/(kg K). Use R_t at hot stations.

$$P = \rho R T, \quad R = c_p\frac{\gamma-1}{\gamma}, \quad a = \sqrt{\gamma R T}, \quad \rho_1 v_1 A_1 = \rho_2 v_2 A_2$$

$$\dot{Q} - \dot{W} = \dot{m}\left[\left(h+\tfrac{v^2}{2}\right)_{out} - \left(h+\tfrac{v^2}{2}\right)_{in}\right]$$

W > 0 extracted (turbine), W < 0 delivered (compressor)

$$s_2 - s_1 = c_p \ln\frac{T_2}{T_1} - R\ln\frac{P_2}{P_1}, \qquad \left.\frac{T_2}{T_1}\right|_s = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$$

$$T_t = T + \frac{v^2}{2c_p}, \quad \frac{T_t}{T} = 1+\frac{\gamma-1}{2}M^2, \quad \frac{P_t}{P} = \left(1+\frac{\gamma-1}{2}M^2\right)^{\frac{\gamma}{\gamma-1}}$$

values in the isentropic flow tables (page 4)

$$\frac{P_{t2}}{P_{t1}} = e^{-\Delta s/R}$$

adiabatic inlet: τ_d = 1 but π_d < 1; normal shock keeps Tt, loses Pt

$$\dot{m} = \frac{P_t A\,\mathrm{MFP}(M)}{\sqrt{RT_t}}, \qquad \mathrm{MFP} = \sqrt{\gamma}M\left(1+\frac{\gamma-1}{2}M^2\right)^{-\frac{\gamma+1}{2(\gamma-1)}}$$

MFP max at M = 1: 0.6673 (γ 1.3), 0.6847 (γ 1.4); oblique flow: A cos(α); Pt in Pa

Station numbering (AS755)

Stn	Location	Stn	Location
0	freestream	4	burner exit / HPT vanes
1	inlet entry	45	HPT exit / LPT entry
2	inlet exit / fan face	5	LPT exit
13	fan exit	6, 16	mixer entry: core, fan duct
25	LPC exit / HPC entry	6A, 7	mixer exit, afterburner exit
3	HPC exit / burner entry	8, 9, 19	throat, core exit, fan exit

Cycle notation and components

$$\tau_r = 1+\frac{\gamma_c-1}{2}M_0^2, \quad \pi_r = \tau_r^{\frac{\gamma_c}{\gamma_c-1}}, \quad \tau_\lambda = \frac{c_{pt}T_{t4}}{c_{pc}T_0}, \quad f = \frac{\dot{m}_f}{\dot{m}_3}$$

$$\pi_d = \pi_{d,\max}\eta_r, \quad \eta_r = 1\ (M_0 \le 1),\ \ 1-0.075(M_0-1)^{1.35}\ (1<M_0<5),$$

$$\eta_d = \frac{(P_{t2}/P_0)^{\frac{\gamma-1}{\gamma}}-1}{T_{t0}/T_0-1}$$

$$\tau_c = \pi_c^{\frac{\gamma_c-1}{\gamma_c e_c}} = 1+\frac{\pi_c^{(\gamma_c-1)/\gamma_c}-1}{\eta_c}, \qquad \dot{W}_c = \dot{m}c_{pc}(T_{t3}-T_{t2})$$

$$\varepsilon = \frac{\gamma-1}{\gamma}:\ \ \eta_c = \frac{\pi_c^\varepsilon-1}{\tau_c-1},\ e_c = \frac{\varepsilon\ln\pi_c}{\ln\tau_c}; \qquad \eta_t = \frac{1-\tau_t}{1-\pi_t^\varepsilon},\ e_t = \frac{\ln\tau_t}{\varepsilon\ln\pi_t}$$

e divides for a compressor (τ = π^(ε/e)), multiplies for a turbine (τ = π^(ε e))

$$f = \frac{c_{pt}T_{t4}-c_{pc}T_{t3}}{\eta_b h_{PR}-c_{pt}T_{t4}} = \frac{\tau_\lambda-\tau_r\tau_c}{\eta_b h_{PR}/(c_{pc}T_0)-\tau_\lambda}, \qquad \pi_b \approx 1-\varepsilon\frac{\gamma}{2}M_b^2$$

jet fuel h_PR ≈ 42.5-42.8 MJ/kg; stoichiometric f ≈ 0.068

$$\tau_t = 1-\frac{\tau_r(\tau_c-1)}{\eta_m(1+f)\tau_\lambda}, \qquad \pi_t = \tau_t^{\frac{\gamma_t}{(\gamma_t-1)e_t}} = \left[1-\frac{1-\tau_t}{\eta_t}\right]^{\frac{\gamma_t}{\gamma_t-1}}$$

$$\frac{P_t}{P^*} = \left(\frac{\gamma+1}{2}\right)^{\frac{\gamma}{\gamma-1}} = 1.832\ (1.3),\ 1.893\ (1.4);$$

$$T^* = \frac{2T_t}{\gamma+1}, \quad P^* = P_t\left[1-\frac{1-T^*/T_t}{\eta_n}\right]^{\frac{\gamma}{\gamma-1}}$$

P* ≥ P0 → choked, P9 = P*, M9 = 1; else matched P9 = P0, M9 < 1

$$T_9 = T_{t5}\left[1-\eta_n\left(1-\left(\frac{P_9}{P_{t5}}\right)^{\frac{\gamma_t-1}{\gamma_t}}\right)\right],$$

$$M_9 = \sqrt{\frac{2}{\gamma_t-1}\left(\frac{T_{t9}}{T_9}-1\right)}, \quad v_9 = \sqrt{2c_{pt}(T_{t5}-T_9)}$$

η_n route refers to Pt5; π_n route uses Pt9 = π_n Pt5 and M9 from Pt9/P9 (do not mix them)

$$\frac{P_{t19}}{P_{19}} = \frac{P_0}{P_{19}}\pi_r\pi_d\pi_f\pi_{fn}, \qquad T_{t19} = T_{t13}$$

civil fan nozzle: convergent and often unchoked at cruise: always do the choked check

Thrust and performance

$$F = \dot{m}_9 v_9 - \dot{m}_0 v_0 + (P_9-P_0)A_9, \qquad A_9 = \frac{\dot{m}_9 R T_9}{P_9 v_9}, \qquad I = PA(1+\gamma M^2)$$

two streams: add $\dot{m}_{19}$ v19 + (P19 - P0) A19; force on a component = I_in - I_out

$$F_{inst} = F - D_{nac} - D_{add}, \qquad F_{adim} = \frac{F}{a_0\dot{m}_0}, \qquad I_{sp} = \frac{F}{\dot{m}_f g} = \frac{1}{g\,\mathrm{TSFC}}$$

$$\mathrm{TSFC} = \frac{\dot{m}_f}{F}, \quad \eta_T = \frac{\Delta E_k}{\dot{m}_f h_{PR}}, \quad \eta_P = \frac{F' v_0}{\Delta E_k}, \quad \eta_O = \eta_T\eta_P = \frac{v_0}{h_{PR}\,\mathrm{TSFC}}$$

TSFC in mg/(N s) = 1e-6 kg/(N s); uninstalled F in the efficiencies

$$\Delta E_k = \tfrac{1}{2}\left(\dot{m}_9 v_9^2 + \dot{m}_{19}v_{19}^2 - \dot{m}_0 v_0^2\right), \qquad \eta_P \approx \frac{2}{1+v_9/v_0}$$

η_P with the FULL thrust (pressure terms included); static (v0 = 0): $\eta_P = \eta_O$ = 0

$$\frac{P_{t9}}{P_9} = \frac{P_0}{P_9}\pi_r\pi_d\pi_c\pi_b\pi_t\pi_n, \qquad M_9 = \sqrt{\frac{2}{\gamma_t-1}\left[\left(\frac{P_{t9}}{P_9}\right)^{\frac{\gamma_t-1}{\gamma_t}}-1\right]}$$

$$\frac{T_9}{T_0} = \frac{\tau_\lambda\tau_t c_{pc}/c_{pt}}{(P_{t9}/P_9)^{(\gamma_t-1)/\gamma_t}}, \qquad \frac{v_9}{a_0} = M_9\sqrt{\frac{\gamma_t R_t T_9}{\gamma_c R_c T_0}}$$

$$\frac{F}{\dot{m}_0} = a_0\left[(1+f)\frac{v_9}{a_0} - M_0 + (1+f)\frac{R_t}{R_c}\frac{T_9/T_0}{v_9/a_0}\frac{1-P_0/P_9}{\gamma_c}\right], \qquad \mathrm{TSFC} = \frac{f}{F/\dot{m}_0}$$

Turbojet in 8 steps: (1) a0, v0, τ_r, π_r (2) η_r, π_d (3) τ_c (4) τ_{lambda}, f (5) τ_t from the power balance, then π_t (6) nozzle: Pt9/P9 → M9, choked check (7) T9/T0, v9/a0 (8) F/$\dot{m}_0$, TSFC, efficiencies.

$$\tau_{c,\mathrm{opt}} = \frac{\sqrt{\tau_\lambda}}{\tau_r}$$

raising Tt4 and π_c raises η_T but also v9, which lowers η_P: the turbofan breaks that coupling

Turbofan and two spools

$$\alpha = \frac{\dot{m}_{sec}}{\dot{m}_{core}}, \quad \dot{m}_{core} = \frac{\dot{m}_0}{1+\alpha}, \quad \dot{m}_f = f\dot{m}_{core}, \quad \mathrm{TSFC} = \frac{f}{(1+\alpha)F/\dot{m}_0}$$

$$\dot{W}_{HPC} = \eta_{m,HP}(1+f)\dot{m}_{core}c_{pt}(T_{t4}-T_{t45})$$

$$\dot{W}_{fan} + \dot{W}_{LPC} = \eta_{m,LP}(1+f)\dot{m}_{core}c_{pt}(T_{t45}-T_{t5}),$$

$$\dot{W}_{fan} = \dot{m}_0 c_{pc}(T_{t13}-T_{t2}), \qquad \dot{W}_{LPC} = \dot{m}_{core}c_{pc}(T_{t25}-T_{t13})$$

$$\mathrm{OPR} = \pi_f\pi_{LPC}\pi_{HPC}, \quad T_{t,out} = T_{t,in}\left[1+\frac{\pi^\varepsilon-1}{\eta}\right],$$

$$\frac{P_{t,out}}{P_{t,in}} = \left[1-\frac{1-T_{t,out}/T_{t,in}}{\eta_t}\right]^{\frac{\gamma_t}{\gamma_t-1}}$$

$$F = \dot{m}_9 v_9 + \dot{m}_{19}v_{19} - \dot{m}_0 v_0 + (P_9-P_0)A_9 + (P_{19}-P_0)A_{19},$$

$$FR = \frac{F_{core}/\dot{m}_{core}}{F_{sec}/\dot{m}_{sec}}$$

$\dot{m}_9$ = (1+f) $\dot{m}_{core}$, $\dot{m}_{19}$ = $\dot{m}_{sec}$

Order: compressions → f → HP balance (Tt45) → LP balance (Tt5) → π from η → nozzle checks → F, TSFC, efficiencies. Per unit core flow everything carries 1/(1+α) when written per $\dot{m}_0$.

Mixed exhaust, mixer and afterburner

Long-duct turbofan: fan duct π_{fd} = Pt16/Pt13 = 0.98-0.99, τ_{fd} = 1. Kutta condition at the splitter: P6 = P16; design approximation Pt6 ≈ Pt16 closes the cycle, so the bypass ratio becomes an OUTPUT.

$$\pi_t = \frac{\pi_{fd}\pi_f}{\pi_b\pi_c}, \quad \tau_t = \pi_t^{\frac{(\gamma_t-1)e_t}{\gamma_t}}, \quad \alpha = \frac{\eta_m(1+f)\tau_\lambda(1-\tau_t)-\tau_r(\tau_c-1)}{\tau_r(\tau_f-1)}$$

negative or huge α: π_f, π_c, Tt4 cannot be matched; f does not depend on α

$$\frac{P_{t6}}{P_0} = \pi_r\pi_d\pi_c\pi_b\pi_t, \quad \frac{T_{t6}}{T_0} = \tau_\lambda\tau_t\frac{c_{pc}}{c_{pt}}; \qquad \frac{P_{t16}}{P_0} = \pi_r\pi_d\pi_f\pi_{fd}, \quad \frac{T_{t16}}{T_0} = \tau_r\tau_f$$

$$\alpha' = \frac{\dot{m}_{16}}{\dot{m}_6} = \frac{\alpha}{1+f}, \quad c_{p6A} = \frac{c_{p6}+\alpha' c_{p16}}{1+\alpha'},$$

$$R_{6A} = \frac{R_6+\alpha' R_{16}}{1+\alpha'}, \quad \gamma_{6A} = \frac{c_{p6A}}{c_{p6A}-R_{6A}}$$

mass-average cp and R, then derive γ (never average γ)

$$c_{p6A}T_{t6A} = \frac{c_{p6}T_{t6} + \alpha' c_{p16}T_{t16}}{1+\alpha'},$$

$$M_{16} = \sqrt{\frac{2}{\gamma_{16}-1}\left[\left(\frac{P_{t16}}{P_{t6}}\left(1+\frac{\gamma_6-1}{2}M_6^2\right)^{\frac{\gamma_6}{\gamma_6-1}}\right)^{\frac{\gamma_{16}-1}{\gamma_{16}}} - 1\right]}$$

M6 = 0.4-0.5 is a design choice

$$\frac{A_{16}}{A_6} = \alpha'\sqrt{\frac{R_{16}T_{t16}}{R_6T_{t6}}}\frac{P_{t6}}{P_{t16}}\frac{\mathrm{MFP}(\gamma_6, M_6)}{\mathrm{MFP}(\gamma_{16}, M_{16})}, \qquad A_{6A} = A_6 + A_{16}$$

$$P_{6A}A_{6A}(1+\gamma_{6A}M_{6A}^2) = P_6A_6\left[(1+\gamma_6M_6^2) + \frac{A_{16}}{A_6}(1+\gamma_{16}M_{16}^2)\right]$$

solve for M6A (iterate from M6A = M6, subsonic root)

$$\pi_M = \frac{P_{t6A}}{P_{t6}} = \pi_{M\max}(1+\alpha')\sqrt{\frac{R_{6A}T_{t6A}}{R_6T_{t6}}}\frac{A_6}{A_{6A}}\frac{\mathrm{MFP}(\gamma_6, M_6)}{\mathrm{MFP}(\gamma_{6A}, M_{6A})}$$

$\pi_{M,max} \approx 0.97$ (friction); the rest is mixing loss

$$f_{AB} = (1+f+\alpha)\frac{c_{pAB}T_{t7} - c_{p6A}T_{t6A}}{\eta_{AB}h_{PR} - c_{pAB}T_{t7}}, \qquad \pi_{AB} = \frac{P_{t7}}{P_{t6A}}, \quad f_{tot} = f + f_{AB}$$

fuel fractions referred to the core flow; turbojet: α = 0 and station 6 for 6A; dry: $\pi_{AB} = \tau_{AB} = 1$, f_AB = 0

$$\mathrm{MFP}(\gamma_7, M_7) = \frac{1+\alpha+f_{tot}}{1+\alpha+f}\,\mathrm{MFP}(\gamma_{6A}, M_{6A})\sqrt{\frac{R_7T_{t7}}{R_{6A}T_{t6A}}}\frac{P_{t6A}}{P_{t7}}$$

thermal choking if the right side > MFP(1) = 0.667 (γ 1.3): keep the afterburner inlet M ≈ 0.2-0.3

$$\frac{P_{t9}}{P_9} = \frac{P_0}{P_9}\pi_r\pi_d\pi_c\pi_b\pi_t\pi_M\pi_{AB}\pi_n, \qquad \mathrm{TSFC} = \frac{f_{tot}}{(1+\alpha)F/\dot m_0}$$

$$\frac{F}{\dot m_0} = \frac{a_0}{1+\alpha}\left[(1+f_{tot}+\alpha)\frac{v_9}{a_0} - (1+\alpha)M_0 + (1+f_{tot}+\alpha)\frac{R_{6A}}{R_c}\frac{T_9/T_0}{v_9/a_0}\frac{1-P_0/P_9}{\gamma_c}\right]$$

lighting the afterburner ≈ doubles TSFC for ≈ 50 % more thrust (v9 grows ≈ sqrt(Tt7))

$$\frac{F_{g,mixed}}{F_{g,unmixed}} = \frac{\sqrt{(1+\alpha)(T_h/T_c+\alpha)}}{\sqrt{T_h/T_c}+\alpha}$$

2-7 %, largest near α ≈ 2 and for hot cores; high-bypass civil engines skip it for duct weight and drag

Turboshaft

$$\tau_{tH} = 1 - \frac{\tau_r(\tau_c-1)}{\eta_{mH}(1+f)\tau_\lambda}, \qquad \frac{\dot W_{shaft}}{\dot m_0} = \eta_{mL}(1+f)c_{pc}T_0\tau_\lambda\tau_{tH}(1-\tau_{tL})$$

$$\dot W_{eq} = \dot W_{shaft} + Fv_0, \quad \mathrm{PSFC} = \frac{f}{\dot W_{shaft}/\dot m_0}, \quad \eta_T = \frac{\dot W_{shaft} + \Delta E_k}{\dot m_f h_{PR}}$$

residual jet: Pt9/P9 = (P0/P9) $\pi_r\,\pi_d\,\pi_c\,\pi_b\,\pi_{tH}\,\pi_{tL}\,\pi_n$; η_P not meaningful without the rotor

Propellers, turboprop, open rotor

$$v_p = \frac{v_0+v_e}{2}, \quad a = \frac{v_p - v_0}{v_0}, \quad F = 2\dot m a v_0,$$

$$\dot m = \rho_0 A_D v_p, \quad \dot W_D = Fv_p, \quad \eta_P = \frac{1}{1+a}$$

$$v_p = \frac{v_0 + \sqrt{v_0^2 + 2F/(\rho_0 A_D)}}{2}, \quad \eta_{prop} = \eta_L\eta_P,$$

$$\eta_L = \frac{\dot W_D}{\dot W_{prop}} = \eta_{prop}(1+a) \le 1$$

$\eta_L > 1$: disc too small for the assumed η_{prop}; v_prop = ve = 2 vp - v0

$$\eta_{prop} = 10M_0\eta_{\max}\ (M_0 \le 0.1), \quad \eta_{\max}\ (0.1 < M_0 \le 0.7),$$

$$\eta_{\max}\left[1 - \frac{M_0 - 0.7}{3}\right]\ (0.7 < M_0 \le 0.85)$$

$\eta_{prop,max}$ 0.80-0.90; chart $\eta_{prop} = \eta_L\,\eta_P$, no gearbox

$$F_{static} = (2\rho_0 A_D)^{1/3}(FM\,\dot W_{prop})^{2/3},$$

$$FM = \frac{F_{static}^{3/2}}{\sqrt{2\rho_0 A_D}\,\dot W_{prop}} \approx 0.70\text{-}0.80, \quad v_s = \sqrt{\frac{F}{2\rho_0 A_D}}$$

x2 power → +59 % static thrust, x2 disc area → +26 %

$$F\big|_{M_0=0} = \frac{10\eta_{\max}\dot W_{prop}}{a_0}, \qquad FM_{implied} = \left(\frac{10\eta_{\max}}{a_0}\right)^{3/2}\sqrt{\frac{\dot W_{prop}}{2\rho_0 A_D}}$$

FM_implied > 1 is impossible: then use the FM formula with FM = 0.70-0.80

$$\dot W_{prop} = \eta_g(\eta_{m,L}\dot W_{LPT} - \dot W_{LPC}),$$

$$F_{prop} = \frac{\eta_{prop}\dot W_{prop}}{v_0}, \quad \dot W_D = (1+a)\eta_{prop}\dot W_{prop}$$

$$\beta = \frac{\dot m_{prop}}{\dot m_{jet}}, \ \dot m_{jet} = \frac{\dot m_0}{1+\beta},$$

$$\frac{\dot W_{prop}}{\dot m_0} = \frac{\eta_g c_{pc}T_0}{1+\beta}\left[\eta_{mL}(1+f)\tau_\lambda\tau_{tH}(1-\tau_{tL}) - \tau_r(\tau_{cL}-1)\right]$$

τ_{tH} from the HP balance with $\tau_r\,\tau_{cL}$ (τ_{cH} - 1) on top

$$v_0\frac{F}{\dot m_0} = \eta_{prop}\frac{\dot W_{prop}}{\dot m_0} + \frac{\gamma_c - 1}{1+\beta}M_0\left[(1+f)\frac{v_9}{a_0} - M_0\right]c_{pc}T_0$$

$$\dot m_{prop} = \rho_0 A_D v_p, \quad \Delta E_k = \tfrac{1}{2}\dot m_9 v_9^2 - \tfrac{1}{2}\dot m_{jet}v_0^2 + \dot W_D,$$

$$\mathrm{TSFC} = \frac{f}{(1+\beta)F/\dot m_0}, \quad \mathrm{PSFC} = \frac{f}{(1+\beta)\dot W_{prop}/\dot m_0}$$

Mattingly uses W_prop instead of W_D: same η_O, lower η_P and higher η_T

$$J = \frac{v_0}{ND}, \quad C_P = \frac{\dot W_{prop}}{\rho_0 N^3 D^5}, \quad \eta_{prop} = \frac{C_F}{C_P}J, \quad M_{tip} = \frac{\sqrt{v_0^2 + (\pi ND)^2}}{a_0}$$

C_P = 1.76327e8 W[kW]/(σ N[rpm]^3 D^5); drag divergence near M0 0.7, swept thin blades reach ≈ 0.8

$$\tau_{tL}^* = \frac{\tau_{tH}^{-1/e_{tH}}}{\Pi}\tau_{tL}^{-\frac{1-e_{tL}}{e_{tL}}} + A\left[1 + \frac{1-e_{tL}}{e_{tL}}\frac{\tau_{tH}^{-1/e_{tH}}\tau_{tL}^{-1/e_{tL}}}{\Pi}\right]^2$$

$$A = \frac{(\gamma_c - 1)M_0^2/2}{\tau_\lambda\tau_{tH}(\eta_{prop}\eta_g\eta_{mL})^2}, \qquad \Pi = (\pi_r\pi_d\pi_{cL}\pi_{cH}\pi_b\pi_n)^{\frac{\gamma_t-1}{\gamma_t}}$$

faster flight or worse propeller → larger τ_{tL}*: more energy left in the jet

$$\alpha_1 = \frac{\dot m_{prop}}{\dot m_{jet}}, \quad \alpha_2 = \frac{\dot m_{sec}}{\dot m_{core}},$$

$$\mathrm{TSFC} = \frac{f}{(1+\alpha_1)(1+\alpha_2)F/\dot m_0}, \quad A_D = \frac{\pi D^2}{4} - \dot m_{45}\frac{\sqrt{RT_{t45}}}{P_{t45}\mathrm{MFP}_{45}}$$

tractor propeller: $A_D \approx \pi\,D^2/4$ (core streamtube neglected)

Engine sizing

$$\dot m_0 = \frac{F_{inst} + D_{nac} + D_{add}}{F/\dot m_0}, \qquad A_i = \frac{\dot m_i\sqrt{R_iT_{ti}}}{P_{ti}\,\mathrm{MFP}(\gamma_i, M_i)}$$

$\dot m_i$: $\dot m_0$ at 0-3; (1+f) $\dot m_0$/(1+α) at core 4-9; α $\dot m_0$/(1+α) at fan 13-19; (1+f+α) $\dot m_0$/(1+α) at 6A-9 (mixed). R_c, γ_c cold, R_t, γ_t hot. C-D nozzle: throat 8 and exit 9 are different areas.

Turbomachinery: Euler and conventions

$$\tau_A = \dot m(r_e v_e - r_i v_i), \qquad h_{te} - h_{ti} = \omega(r_e v_e - r_i v_i)$$

$$\text{compressor: } c_p\Delta T_t = U(v_2 - v_1),$$

$$\text{turbine: } c_p\Delta T_t = U(v_2 + v_3), \quad U = \omega r_m$$

$$\vec V = \vec V_R + \vec U, \quad v = u\tan\alpha, \quad v_R = u\tan\beta, \quad r_m = \frac{r_t + r_h}{2}, \quad \omega = \frac{2\pi N}{60}$$

compressor: rotor 1-2 then stator 2-3, v + v_R = U; turbine: nozzle 1-2 then rotor 2-3, v2R = v2 - U, v3R = v3 + U (v3 positive AGAINST U)

P, T, ρ are frame-invariant; Tt, Pt, M are not. Rotor: Tt,R constant (same radius). Stator: Tt constant.

Compressor stage

$$c_p\Delta T_t = Uu(\tan\beta_1 - \tan\beta_2) = Uu(\tan\alpha_2 - \tan\alpha_1)$$

$$T_{t1R} = T_1 + \frac{V_{1R}^2}{2c_p} = T_{t2R}, \quad P_{t2R} = L_R P_{t1R},$$

$$P_{t3} = L_S P_{t2}, \quad L_R = 1 - \phi_{cr}\frac{\gamma P_1 M_{1R}^2}{2P_{t1R}}$$

$$R_c = \frac{T_2 - T_1}{T_3 - T_1}, \quad \Psi = \frac{c_p\Delta T_t}{U^2},$$

$$\Phi = \frac{u_1}{U} = \frac{1}{\tan\alpha_1 + \tan\beta_1}, \quad \frac{\Psi}{\Phi} = \tan\beta_1 - \frac{u_2}{u_1}\tan\beta_2$$

$$\eta_{cs} = \frac{\pi_{cs}^{\varepsilon} - 1}{\tau_{cs} - 1}, \qquad e_{cs} = \frac{\gamma - 1}{\gamma}\frac{\ln(P_{t3}/P_{t1})}{\ln(T_{t3}/T_{t1})}, \qquad M_b = \frac{U}{a_1}$$

$$D = 1 - \frac{V_e}{V_i} + \frac{|v_i - v_e|}{2\sigma V_i} \le 0.6,$$

$$D_r = 1 - \frac{V_{2R}}{V_{1R}} + \frac{|v_{1R} - v_{2R}|}{2\sigma V_{1R}}, \quad D_s = 1 - \frac{V_3}{V_2} + \frac{|v_2 - v_3|}{2\sigma V_2}$$

de Haller Ve/Vi ≥ 0.72 (rough)

$$\sigma = \frac{c}{s}, \quad s = \frac{2\pi r_m}{n_b}, \quad i = \alpha_i - \gamma_i, \quad \delta = \alpha_e - \gamma_e, \quad \delta_c = \frac{\gamma_i - \gamma_e}{4\sqrt{\sigma}}, \quad \delta_t = \frac{\gamma_i + \gamma_e}{8\sqrt{\sigma}}$$

$$\gamma_e = \frac{4\alpha_e\sqrt{\sigma} - \gamma_i}{4\sqrt{\sigma} - 1}$$

$$\text{repeating stage: } v_1 + v_2 = U, \quad c_p \Delta T_t = V_2^2 - V_1^2,$$

$$\frac{\omega r}{V_1} = \cos\alpha_1(\tan\alpha_1 + \tan\alpha_2)$$

V1R = V2, V2R = V1: solve one triangle; T3 = Tt3 - V1^2/(2cp)

$$\tau_{cs} = 1 + \frac{(\gamma-1)M_1^2}{1 + \frac{\gamma-1}{2}M_1^2}\left(\frac{\cos^2\alpha_1}{\cos^2\alpha_2} - 1\right),$$

$$\Psi = \frac{\tan\alpha_2 - \tan\alpha_1}{\tan\alpha_1 + \tan\alpha_2}, \quad \Phi = \frac{1}{\tan\alpha_1 + \tan\alpha_2}$$

$$D = 1 - \frac{\cos\alpha_2}{\cos\alpha_1} + \frac{\tan\alpha_2 - \tan\alpha_1}{2\sigma}\cos\alpha_2, \qquad M_3 = M_1\sqrt{\frac{T_1}{T_3}}, \qquad \pi_{cs} = \tau_{cs}^{\frac{\gamma e_c}{\gamma-1}}$$

$$\pi_c = \left[\frac{T_{t1} + n_s\Delta T_t}{T_{t1}}\right]^{\frac{\gamma e_c}{\gamma-1}},$$

$$n_s \approx \frac{T_{t1}\left(\pi_c^{(\gamma-1)/(\gamma e_c)} - 1\right)}{\Delta T_{t,\text{stage}}}, \quad M_{3,n} = M_1\sqrt{\frac{T_1}{T_1 + n_s\Delta T_t}}$$

stage ratios fall along the machine: $\pi_c \neq \pi_{cs}$^n; first stage set by inlet guide vanes

$$\frac{dh_t}{dr} = u\frac{du}{dr} + v\frac{dv}{dr} + \frac{v^2}{r}, \qquad \text{free vortex: } R = 1 - (1 - R_m)\left(\frac{r_m}{r}\right)^2$$

reaction lowest at the hub (turbine: keep R_t > -0.15 there)

$$A = \frac{\dot m\sqrt{RT_t}}{P_t\cos\alpha\,\mathrm{MFP}(M)}, \quad r_t = \sqrt{\frac{A}{\pi\left[1 - (r_h/r_t)^2\right]}},$$

$$h = \frac{A}{2\pi r_m}, \quad \omega = \frac{U_t}{r_t}, \; N = \frac{30\omega}{\pi}$$

hub/tip 0.60-0.75 at HPC inlet, 0.90-0.92 at exit; Pt in Pa

Compressor design range	Fan / LPC	HPC
pressure ratio: 1 / 2 / 3 stages	1.5-2.0 / 2.0-3.5 / 3.5-4.5	-
inlet corrected ṁ per area, kg/(s m^2)	195-205	175-185
tip speed (m/s)	427-457	rim at exit 396-457
diffusion factor D	0.50-0.55	0.50-0.55
Φ / Ψ	-	0.45-0.55 / 0.30-0.35
first stage: hub/tip, axial M, U_tip	0.6-0.75, 0.48-0.6	350-460 m/s

Turbine stage

$$T_{t2} = T_{t1}, \quad P_{t2} = L_S P_{t1}, \quad T_{t3R} = T_{t2R}, \quad P_{t3R} = L_R P_{t2R},$$

$$L = \frac{1}{1 + \phi_t\left[1 - (T/T_t)^{\frac{\gamma}{\gamma-1}}\right]}$$

$$\Psi = \frac{c_p\Delta T_t}{U^2} = \frac{u_2}{U}\left(\tan\alpha_2 + \frac{u_3}{u_2}\tan\alpha_3\right) = \frac{u_2}{U}\left(\tan\beta_2 + \frac{u_3}{u_2}\tan\beta_3\right),$$

$$\Phi = \frac{u_2}{U} = \frac{1}{\tan\alpha_2 - \tan\beta_2}$$

$$\text{const. } u: \; \frac{\Psi}{\Phi} = \tan\beta_2 + \tan\beta_3, \quad \frac{R_t}{\Phi} = \frac{\tan\beta_3 - \tan\beta_2}{2}, \quad R_t = \frac{T_2 - T_3}{T_{t1} - T_{t3}}$$

$$R_t = 0: \; \Psi = 2\Phi\tan\beta_2 \; (\beta_3 = \beta_2);$$

$$R_t = 0.5: \; \Psi = 2\Phi\tan\alpha_2 - 1 \; (\beta_3 = \alpha_2); \quad \alpha_3 = 0: \; \Psi = 2(1 - R_t)$$

zero swirl + 0 %: Ψ = 2, v2 = 2U, v2R = v3R = U; zero swirl + 50 %: Ψ = 1; Ψ > 2 means R_t < 0

$$\Phi = \frac{1}{\tan\alpha_2 - \tan\beta_2}, \qquad U = \sqrt{\frac{c_p\Delta T_t}{\Psi}}, \qquad u = \Phi U$$

map axes: turning $\beta_2 + \beta_3$ vs β_2; pick the LARGEST Ψ/Φ with $\alpha_2 \le 70°$ (and Zweifel)

$$\eta_{ts} = \frac{1 - T_{t3}/T_{t1}}{1 - (P_{t3}/P_{t1})^{(\gamma-1)/\gamma}}, \qquad e_{ts} = \frac{\gamma}{\gamma-1}\frac{\ln(T_{t3}/T_{t1})}{\ln(P_{t3}/P_{t1})}$$

$$\alpha_3 = 0: \; \sin\alpha_2 = \frac{\Psi U}{V_2}, \quad R_t = 1 - \frac{1}{2\Psi}\left(\frac{V_2}{U}\right)^2\left[1 - \left(\frac{V_3}{V_2}\right)^2\right],$$

$$T_3 = T_2 - R_t(T_{t1} - T_{t3}), \quad M_3 = M_2\frac{V_3}{V_2}\sqrt{\frac{T_2}{T_3}}$$

$$M_2\sin\alpha_2 = 2M_T(1 - R_t), \; M_T = \frac{U}{a_2};$$

$$M_{2,opt} = \sqrt{\frac{2}{\gamma-1}\left[\left(\frac{2\left(1 + \frac{\gamma-1}{2}M_1^2\right)^{\frac{\gamma+1}{\gamma-1}}}{(\gamma+1)M_1^2}\right)^{\frac{\gamma-1}{2}} - 1\right]}$$

γ 1.33, M_T 0.5: R_t = 0 → M1 ≈ 0.40, M2 ≈ 1.48; R_t = 0.5 → M1 ≈ 0.65, M2 ≈ 1.14

$$Z = \frac{F_t}{F_{t\max}} = \frac{2s}{c_x}\cos^2\alpha_e\left(\tan\alpha_i + \frac{u_e}{u_i}\tan\alpha_e\right)\left(\frac{u_i}{u_e}\right)^2,$$

$$Z_r = 2\frac{s}{c_x}\cos^2\beta_3(\tan\beta_2 + \tan\beta_3)$$

design Z 0.8-1.0; a limit may be given on Z_r c_x/s instead

$$F_t = \rho u_i s(v_i + v_e) = \rho u_i^2 s\left(\tan\alpha_i + \frac{u_e}{u_i}\tan\alpha_e\right),$$

$$F_{t\max} = (P_{ti} - P_e)c_x = \frac{\rho u_e^2 c_x}{2\cos^2\alpha_e}$$

$$\sigma_c \approx \frac{\rho\omega^2 A}{4\pi}\left(1 + \frac{A_t}{A_h}\right) \;\Leftrightarrow\; AN^2 = \frac{3600\times10^3}{\pi(1 + A_t/A_h)}\frac{\sigma_c}{\rho}$$

N in rpm, A in m^2, σ_c/ρ in MPa/(g/cm^3) = 1000 m^2/s^2; too high: lower N, change taper, add a stage

$$\mathrm{DSF} = \frac{(\omega r_r)^2\rho}{2\sigma_d}, \qquad \sigma_{tr} = \frac{\alpha E\Delta T}{3}\left(1 - \frac{r}{r_h}\right), \quad \sigma_{t\theta} = \frac{\alpha E\Delta T}{3}\left(1 - \frac{2r}{r_h}\right)$$

cooled blades allow A N^2 ≈ 4-5e10 in^2 rpm^2

$$T_{t3}^{(n)} = T_{t1}^{(1)} - n_s\Delta T_t$$

only the first nozzle chokes and meters the engine flow

Choking: first nozzle M2 > 1 and rotor exit M3R < 0.9 (a 50 % first stage, Ψ = 1, cannot meter the flow). Sanity: r_m = U/ω and h = A/(2 π r_m) must be realistic; typical LPT U 350-460 m/s.

Fast solving 1: two-spool turbofan and its propeller versions

Turbofan recipe: (1) a0, v0, Tt0, Pt0; Pt2 = π_d Pt0. (2) Tt13, Tt25, Tt3 from π and η, π_{HPC} = OPR/(π_f π_{LPC}); Pt by multiplying the π. (3) f; Pt4 = π_b Pt3. (4) Tt45, Tt5 from the shortcuts below; Pt45, Pt5 from η_t. (5) each nozzle: P* ≥ P0? → T9, P9, v9, A9. (6) powers (kW), F, TSFC, dEk, efficiencies.

$$v_0 = M_0\sqrt{\gamma_c R_c T_0}, \quad T_{t0} = T_0(1 + 0.2M_0^2), \quad P_{t0} = P_0\left(\frac{T_{t0}}{T_0}\right)^{3.5}, \quad \rho_0 = \frac{P_0}{R_c T_0}$$

$$T_{t45} = T_{t4} - \frac{c_{pc}(T_{t3} - T_{t25})}{\eta_{mH}(1 + f)c_{pt}},$$

$$T_{t5} = T_{t45} - \frac{c_{pc}\left[(1 + \alpha)(T_{t13} - T_{t2}) + (T_{t25} - T_{t13})\right]}{\eta_{mL}(1 + f)c_{pt}}$$

$$\tau_t = 1 - \eta_t\left(1 - \pi_t^{\frac{\gamma_t - 1}{\gamma_t}}\right) \;\Leftrightarrow\; \pi_t = \left[1 - \frac{1 - \tau_t}{\eta_t}\right]^{\frac{\gamma_t}{\gamma_t - 1}} \qquad \left(\tfrac{\gamma_t}{\gamma_t - 1} = 4.333\right)$$

$$\text{choked: } T^* = \frac{2T_t}{\gamma + 1}, \; v_9 = \sqrt{\gamma RT^*}, \; P_9 = P^*;$$

$$\text{matched: } P_9 = P_0, \; v_9 = \sqrt{2c_p(T_t - T_9)}; \quad A_9 = \frac{\dot m_9 RT_9}{P_9 v_9}$$

T*/Tt = 0.8696 (γ 1.3), 0.8333 (γ 1.4)

Propfan / open rotor recipe (fan replaced by a propeller): $\dot m_{jet} = \dot m_{core}$; Tt2 → LPC → HPC with the same π_{LPC} and π_{HPC}; f; Tt45 from the HP balance; π_{LPT} kept from the turbofan design, so $\tau_{LPT} = 1 - \eta_{LPT}(1 - \pi_{LPT}^{\varepsilon})$ and Tt5 = τ_{LPT} Tt45; W_LPT = (1+f) $\dot m_{core}$ cp_t (Tt45 - Tt5); W_prop = η_g (η_m W_LPT - W_LPC); core nozzle as before; F_jet = $\dot m_9$ v9 - $\dot m_{core}$ v0 + (P9 - P0) A9.

$$\text{flight: } F_{prop} = \frac{\eta_{prop}\dot W_{prop}}{v_0} \to v_p \to a, \; \dot W_D = F_{prop}v_p, \; \eta_L;$$

$$\text{static: } F_{prop} = (2\rho_0 A_D)^{1/3}(FM\,\dot W_{prop})^{2/3}, \; v_p = \sqrt{\frac{F_{prop}}{2\rho_0 A_D}}$$

at M0 0.8: $\eta_{prop} = \eta_{max}$ (1 - 0.1/3); tractor propeller: A_D = π D^2/4

$$\dot m_{prop} = \rho_0 A_D v_p, \quad \text{bypass} = \frac{\dot m_{prop}}{\dot m_{core}},$$

$$\dot m_0 = \dot m_{core} + \dot m_{prop}, \quad \Delta E_k = \tfrac{1}{2}\left(\dot m_9 v_9^2 - \dot m_{core}v_0^2\right) + \dot W_D$$

Fast solving 2: repeating compressor stage

Given N and r_m: U = ω r_m; the design chart at α_1 gives ω r/V1 and α_2 → V1, M1 = V1/α_1. Given π_c, n_s, e_c: take dTt below, pick M1 (0.4-0.6) on the chart → α_1, α_2; T1, P1 from Tt1, Pt1, M1; V1 = M1 sqrt(γ R T1). Then the same chain: triangle, rotor in the relative frame, stator in the absolute frame, stage figures.

$$u = V_1 \cos\alpha_1, \quad \tan\alpha_2 = \frac{U}{u} - \tan\alpha_1, \quad V_2 = \frac{u}{\cos\alpha_2},$$

$$V_{1R} = V_2,\ V_{2R} = V_1, \quad \Delta T_t = \frac{V_2^2 - V_1^2}{c_p}$$

$$T_{t1R} = T_1 + \frac{V_2^2}{2c_p} = T_{t2R}, \quad T_2 = T_{t1R} - \frac{V_1^2}{2c_p},$$

$$T_{t2} = T_{t1} + \Delta T_t, \quad T_3 = T_{t2} - \frac{V_1^2}{2c_p}, \quad T_{t3R} = T_3 + \frac{V_2^2}{2c_p}$$

$$P_{t1R} = P_1 \left(\frac{T_{t1R}}{T_1}\right)^{3.5}, \quad P_{t2R} = P_{t1R} - \phi_{cr} \tfrac{1}{2}\rho_1 V_2^2,$$

$$P_2 = P_{t2R} \left(\frac{T_2}{T_{t1R}}\right)^{3.5}, \quad P_{t2} = P_2 \left(\frac{T_{t2}}{T_2}\right)^{3.5}$$

rho1 = P1/(R T1); same as L_R = 1 - φ γ P M^2/(2 Pt)

$$P_{t3} = P_{t2} - \phi_{cs} \tfrac{1}{2}\rho_2 V_2^2, \quad P_3 = P_{t3} \left(\frac{T_3}{T_{t2}}\right)^{3.5}, \quad \pi_{cs} = \frac{P_{t3}}{P_{t1}},$$

$$\eta_{cs} = \frac{\pi_{cs}^{0.2857} - 1}{\Delta T_t / T_{t1}}, \quad e_c = \frac{0.2857 \ln \pi_{cs}}{\ln(1 + \Delta T_t / T_{t1})}$$

rho2 = P2/(R T2); R_c = (T2 - T1)/(T3 - T1) = 0.5 for the repeating stage

$$\Delta T_t = \frac{T_{t1}}{n_s} \left(\pi_c^{\frac{\gamma-1}{\gamma e_c}} - 1\right), \quad \pi_{cs} = \left(1 + \frac{\Delta T_t}{T_{t1}}\right)^{\frac{\gamma e_c}{\gamma - 1}},$$

$$\eta_c = \frac{\pi_c^{0.2857} - 1}{n_s \Delta T_t / T_{t1}}, \quad T_{t,exit} = T_{t1} + n_s \Delta T_t$$

check: V2^2 = V1^2 + cp dTt; n stages: π_c = (1 + n dTt/Tt1)^(γ e_c/(γ-1)), not π_{cs}^n

Fast solving 3: axial turbine stage

Tt1, Pt1, Tt3, Pt3 and ṁ from the cycle (HPT: 4 → 45, LPT: 45 → 5). Ψ from the reaction; on the map take the LARGEST Ψ/Φ with $\alpha_2 \le 70°$ and the Zweifel limit; Φ = Ψ/(Ψ/Φ); U = sqrt(cp dTt/Ψ), u = Φ U; angles and speeds below; thermodynamics BACKWARDS from the exit; L_S, η_{ts}; A2 and σ_c/ρ.

$$\alpha_3 = 0,\ u \text{ const.}: \quad \tan\alpha_2 = \frac{\Psi}{\Phi}, \quad \tan\beta_2 = \frac{\Psi - 1}{\Phi}, \quad \tan\beta_3 = \frac{1}{\Phi}, \quad R_t = 1 - \frac{\Psi}{2}$$

50 %: Ψ = 1, $\beta_2 = 0$, $\beta_3 = \alpha_2$; 0 %: Ψ = 2, $\tan\beta_2 = \tan\beta_3 = 1/\Phi$, $\tan\alpha_2 = 2/\Phi$

$$V_2 = \frac{u}{\cos\alpha_2}, \quad V_{2R} = \frac{u}{\cos\beta_2}, \quad V_3 = u, \quad V_{3R} = \frac{u}{\cos\beta_3} = \sqrt{u^2 + U^2}, \quad v_2 = \Psi U$$

$$\frac{Z_r c_x}{s} = 2\cos^2\beta_3(\tan\beta_2 + \tan\beta_3) = 2\sin 2\beta_3\ (R_t = 0), \quad = \sin 2\beta_3\ (R_t = 0.5)$$

$$T_3 = T_{t3} - \frac{u^2}{2c_p},\ P_3 = P_{t3}\left(\frac{T_3}{T_{t3}}\right)^k,\ T_{t3R} = T_3 + \frac{V_{3R}^2}{2c_p} = T_{t2R},\ P_{t3R} = P_3\left(\frac{T_{t3R}}{T_3}\right)^k,\ P_{t2R} = \frac{P_{t3R}}{L_R}$$

k = γ/(γ-1) = 4.333; R = 285.9

$$T_2 = T_{t1} - \frac{V_2^2}{2c_p}, \quad P_2 = P_{t2R}\left(\frac{T_2}{T_{t2R}}\right)^k,$$

$$P_{t2} = P_2\left(\frac{T_{t1}}{T_2}\right)^k, \quad L_S = \frac{P_{t2}}{P_{t1}}, \quad M_2 = \frac{V_2}{\sqrt{\gamma R T_2}}$$

check Tt2R = T2 + V2R^2/(2cp) equals Tt3R; zero reaction: T3 = T2

$$A_2 = \frac{\dot{m}\sqrt{RT_{t1}}}{P_{t2}\cos\alpha_2\,\mathrm{MFP}(M_2)}, \quad \frac{\sigma_c}{\rho} = \frac{\pi(1 + A_t/A_h)AN^2}{3.6 \times 10^6}\ \left[\frac{\mathrm{MPa}}{\mathrm{g/cm^3}}\right], \quad r_m = \frac{U}{\omega}$$

η_{ts} = (1 - Tt3/Tt1)/(1 - (Pt3/Pt1)^(1/k)) should return the cycle η_t

Component figures of merit (Mattingly)

Component	FoM	Lvl 2	Lvl 3	Lvl 4
diffuser (A / C)	$\pi_{d,max}$	0.95 / 0.90	0.98 / 0.94	0.995 / 0.96
compressor / fan	e_c / e_f	0.84 / 0.82	0.88 / 0.86	0.90 / 0.89
burner	π_b / η_b	0.92 / 0.94	0.94 / 0.99	0.95 / 0.999
turbine uncooled / cooled	e_t	0.85 / 0.83	0.89 / 0.87	0.90 / 0.89
afterburner	π_{AB} / η_{AB}	0.92 / 0.91	0.94 / 0.96	0.95 / 0.99
nozzle (D / F)	π_n	0.97 / 0.93	0.98 / 0.95	0.995 / 0.97
shaft (alone / + PTO)	η_m	0.97 / 0.92	0.99 / 0.95	0.995 / 0.97
max Tt4 / Tt7 (K)		1390 / 1670	1780 / 2000	2000 / 2220

A = subsonic, engines in nacelles; C = supersonic, engine in airframe; D = fixed convergent, F = variable C-D nozzle. Level 3 ≈ 1985-2005 technology.

Isentropic flow tables

M	Tt/T 1.4	Pt/P 1.4	MFP 1.4	A/A* 1.4	Tt/T 1.3	Pt/P 1.3	MFP 1.3
0.10	1.0020	1.0070	0.1176	5.8218	1.0015	1.0065	0.1134
0.20	1.0080	1.0283	0.2311	2.9635	1.0060	1.0263	0.2229
0.30	1.0180	1.0644	0.3365	2.0351	1.0135	1.0598	0.3249
0.40	1.0320	1.1166	0.4306	1.5901	1.0240	1.1082	0.4164
0.50	1.0500	1.1862	0.5111	1.3398	1.0375	1.1730	0.4951
0.60	1.0720	1.2755	0.5763	1.1882	1.0540	1.2560	0.5592
0.70	1.0980	1.3871	0.6257	1.0944	1.0735	1.3598	0.6081
0.80	1.1280	1.5243	0.6595	1.0382	1.0960	1.4877	0.6419
0.90	1.1620	1.6913	0.6787	1.0089	1.1215	1.6436	0.6612
1.00	1.2000	1.8929	0.6847	1.0000	1.1500	1.8324	0.6673
1.20	1.2880	2.4250	0.6645	1.0304	1.2160	2.3337	0.6465
1.50	1.4500	3.6710	0.5822	1.1762	1.3375	3.5259	0.5610
2.00	1.8000	7.8244	0.4058	1.6875	1.6000	7.6651	0.3763

Columns for γ = 1.4 (cold) and 1.3 (hot). MFP = ṁ sqrt(R Tt)/(Pt A), dimensionless (R outside). Choked: Tt/T* = 1.2, Pt/P* = 1.893 (γ 1.4); 1.15, 1.832 (γ 1.3).

ISA standard atmosphere

h (km)	T (K)	P (bar)	ρ (kg/m^3)	σ	a (m/s)
0	288.15	1.0132	1.2250	1.0000	340.3
1	281.65	0.8987	1.1117	0.9075	336.4
2	275.15	0.7950	1.0065	0.8216	332.5
3	268.65	0.7011	0.9091	0.7421	328.6
4	262.15	0.6164	0.8191	0.6687	324.6
5	255.65	0.5402	0.7361	0.6009	320.5
6	249.15	0.4718	0.6597	0.5385	316.4
7	242.65	0.4106	0.5895	0.4812	312.3
8	236.15	0.3560	0.5252	0.4287	308.1
9	229.65	0.3074	0.4664	0.3807	303.8
10	223.15	0.2644	0.4127	0.3369	299.5
11	216.65	0.2263	0.3639	0.2971	295.1
12	216.65	0.1933	0.3108	0.2537	295.1
13	216.65	0.1651	0.2655	0.2167	295.1

T = 288.15 - 6.5 h (h in km) up to 11 km, then 216.65 K; P = 1.01325 (T/288.15)^5.2559 bar below 11 km, P = 0.22632 exp(-0.1577 (h - 11)) above. If the statement gives T0 and P0, use those.

Common slips

Choked ratio exponent is γ/(γ-1) (1.832), not γ/(γ+1). e divides for compressors, multiplies for turbines. Fan work uses ALL the flow $\dot{m}_0$. η_g multiplies BOTH the LPT and LPC terms of W_prop. Mechanical losses are charged to the turbine (W_c = η_m W_t). Hot stations use R_t, γ_t, cp_t. Pt in Pa inside MFP and A. The η_n route refers to Pt5, the π_n route to Pt9. η_P with the full thrust, η_O = v0/(h_PR TSFC) as a check. Propeller: W_D, not W_prop, in dEk; check $\eta_L \le 1$ and FM ≤ 1. Turbine v3 is positive AGAINST U. Round n_s UP.